CS459/659 Privacy, Cryptography, Network and Data Security

Cryptography Basics

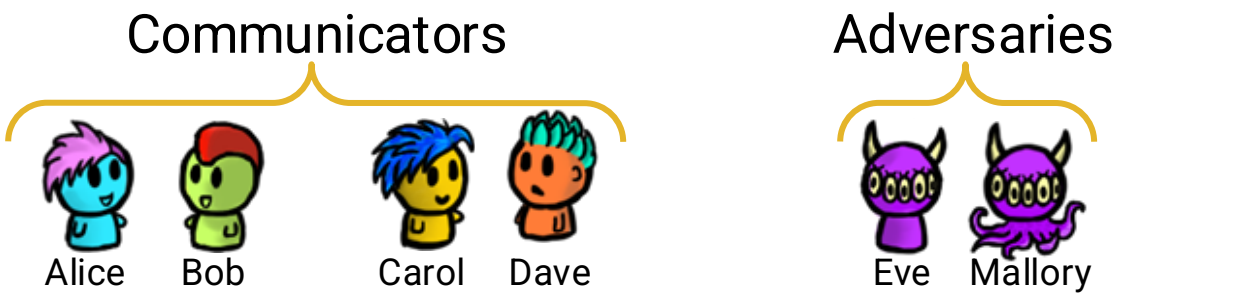
Spring 2026, Tuesday/Thursday 10:00-11:20am

Learning Outcomes

- Identify attack techniques and apply them (cryptanalysis)
- Explain building blocks of modern cryptography
- Explain how modern cryptography properties arose

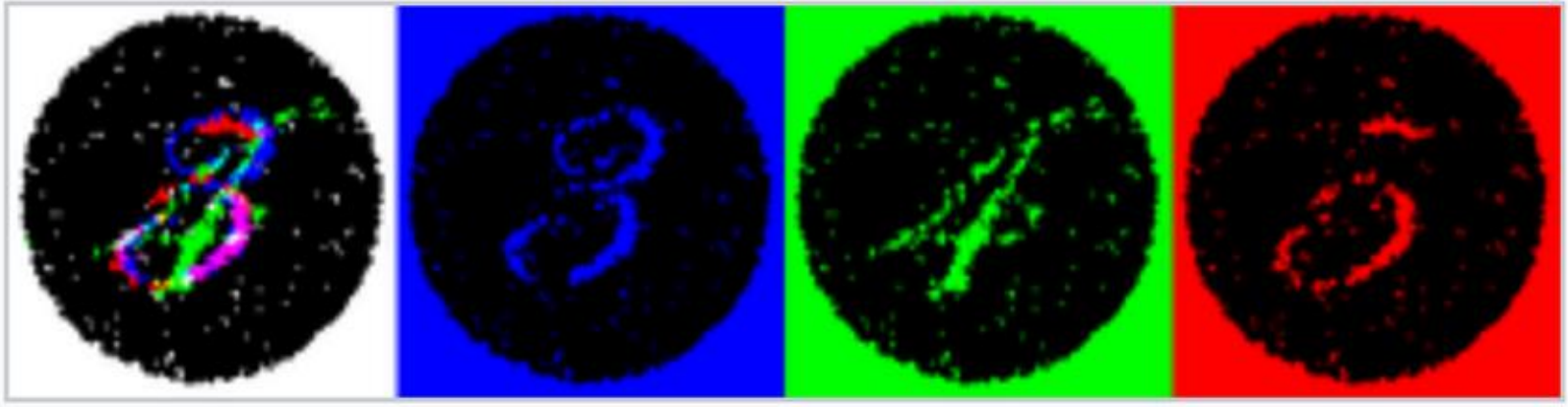
Goal: Basically, know what cryptography tools exist and how to securely use them. Build a foundation of primitives for more complicated “applied cryptography” later.

Cryptography - Writing “secret” messages





Steganography- Secretly “hidden” messages



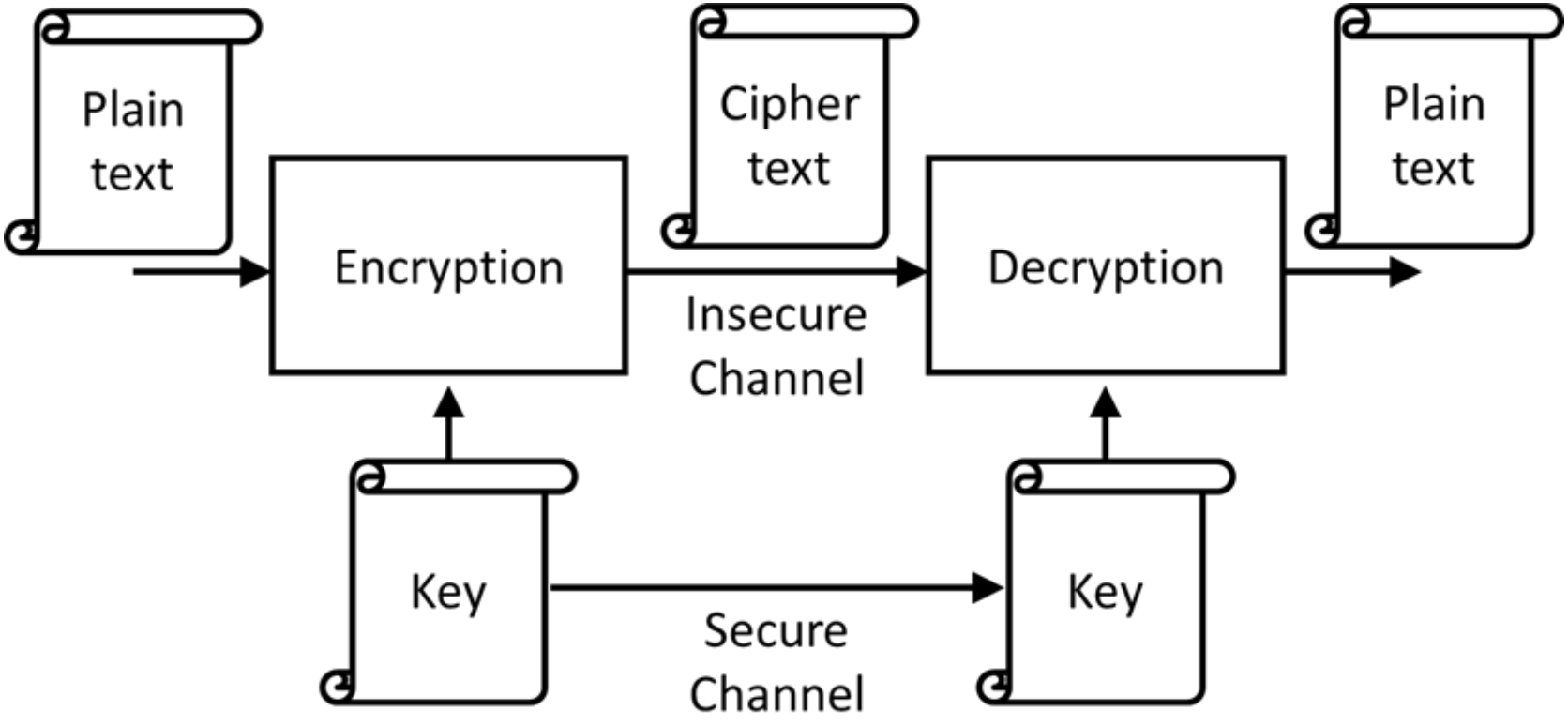
The same image viewed by white, blue, green, and red lights reveals different hidden numbers.

Remember CIA? Different A for Crypto Power

- Confidentiality, prevent Eve **reading** Alice's messages
- Integrity, prevent Mallory from **changing** Alice's messages
- Authenticity, Prevent Mallory from **impersonating** Alice



Cryptography - Path for Secret Messages



Historical Ciphers: Example One

FUBSWRJUDSKB

CRYPTOGRAPHY

Caesar Cipher

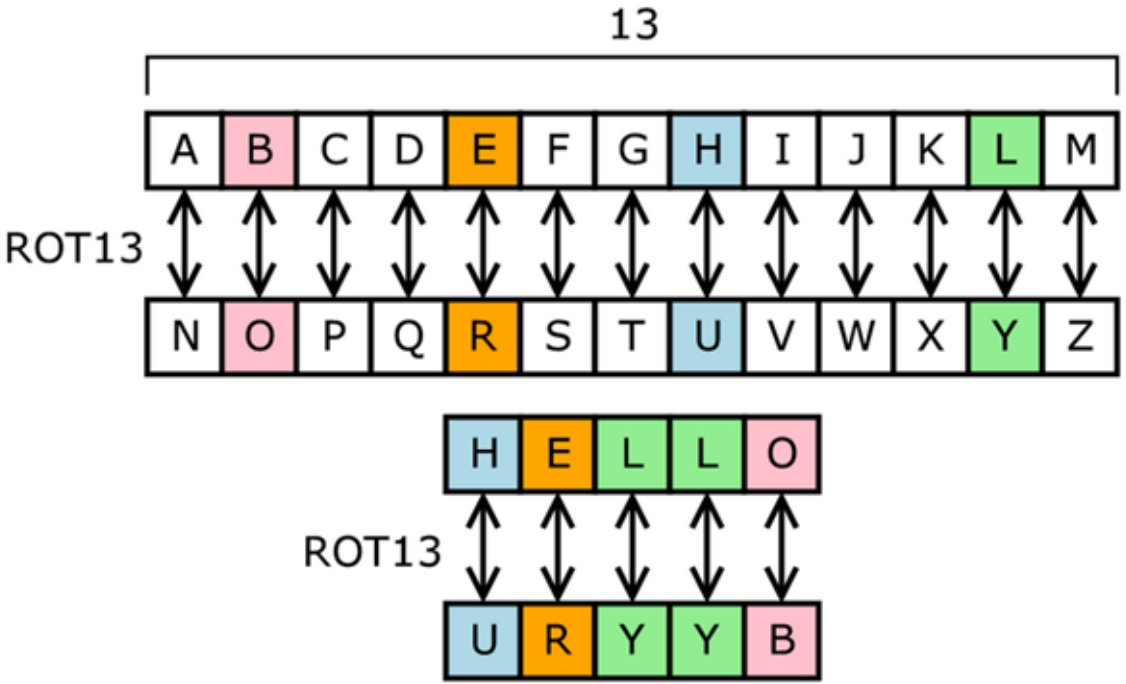


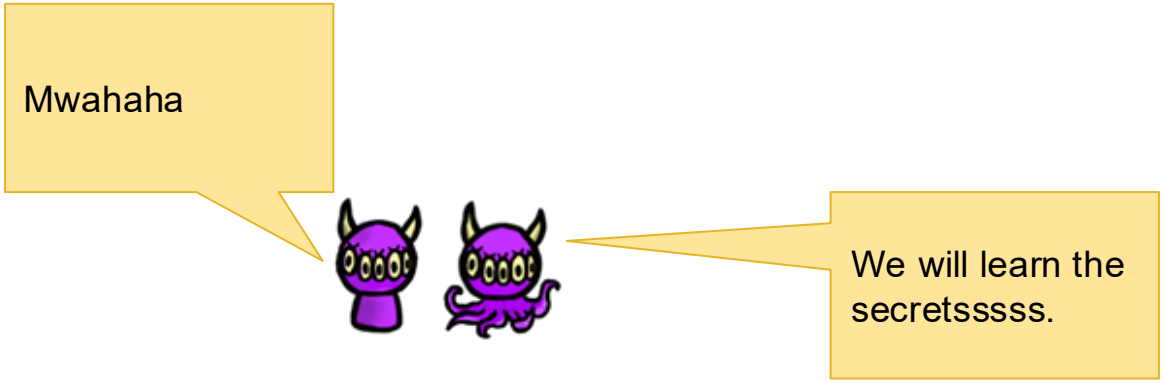
Image source: wikipedia

Shift and Substitution Ciphers

Replace symbols (letters) by others

- Using a rule e.g., y = x + 3 (mod 26), Caesar‘s cipher Key: 3
- Using a table e.g, Key: table

Cryptanalysis - Analyzing “secret” messages

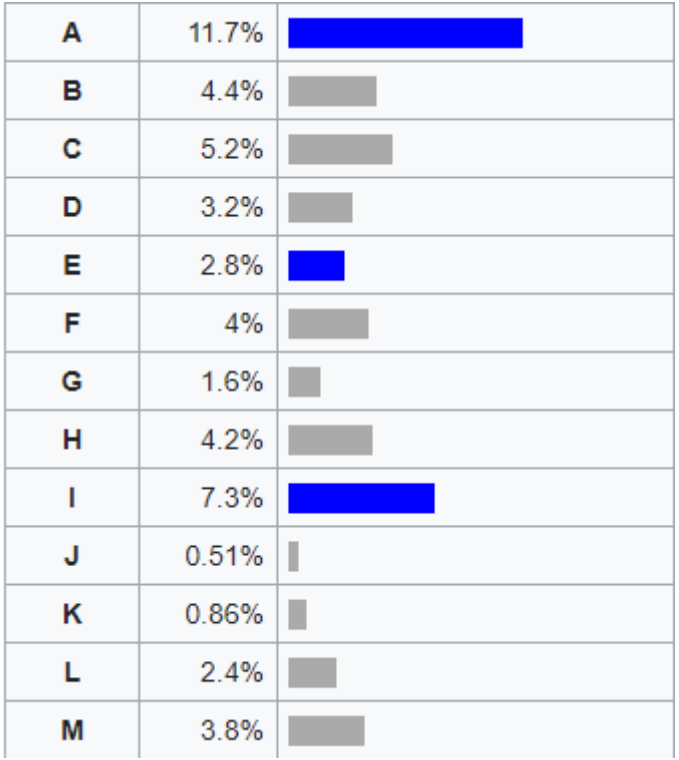


Historical Ciphers: Example Two

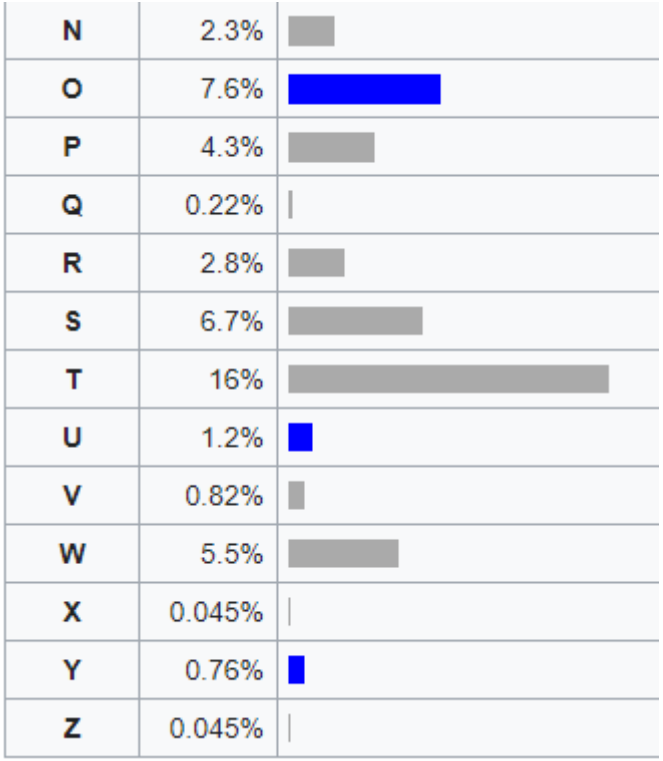
gsrh xlfihv rh zylfg xibkgltizksb uli gsv urihg gsivv dvvph. zmw gsvm zkkorvw xibkgltizksb uli kirezxb zmw hvxfirgb lu wzgz.



English Frequency



A	11.7%
B	4.4%
C	5.2%
D	3.2%
E	2.8%
F	4%
G	1.6%
H	4.2%
I	7.3%
J	0.51%
K	0.86%
L	2.4%
M	3.8%



N	2.3%
O	7.6%
P	4.3%
Q	0.22%
R	2.8%
S	6.7%
T	16%
U	1.2%
V	0.82%
W	5.5%
X	0.045%
Y	0.76%
Z	0.045%

Historical Ciphers: Example Two



gsrh xlfihv rh zylfg xibkgltizksb uli **gs**v urihg **gs**ivv dvvph. zmw **gs**vm zkkorvw xibkgltizksb uli kirezxb zmw hvxfirgb lu wzgz.

Historical Ciphers: Example Two



gsrh xlfihv rh zylfg xibkgltizksb uli **gs**v urihg **gs**ivv dvvph. zmw **gs**vm zkkorvw xibkgltizksb uli kirezxb zmw hvxfirgb lu wzgz.



This course is about cryptography for **th**e first **th**ree weeks. And **th**en applied cryptography for privacy and security of data.

Kerckhoff Principle

The security of a cryptosystem should solely depend on the secrecy of the key, but never on the secrecy of the algorithms.

Historical Ciphers: Example Three

LECTURE SECURITY AND CRYPTOGRAPHY I



LENGECDRCUCATRRPUIYHRTPYEYTISAO

Historical Ciphers: Example Three

LECTURES

ECURITYA

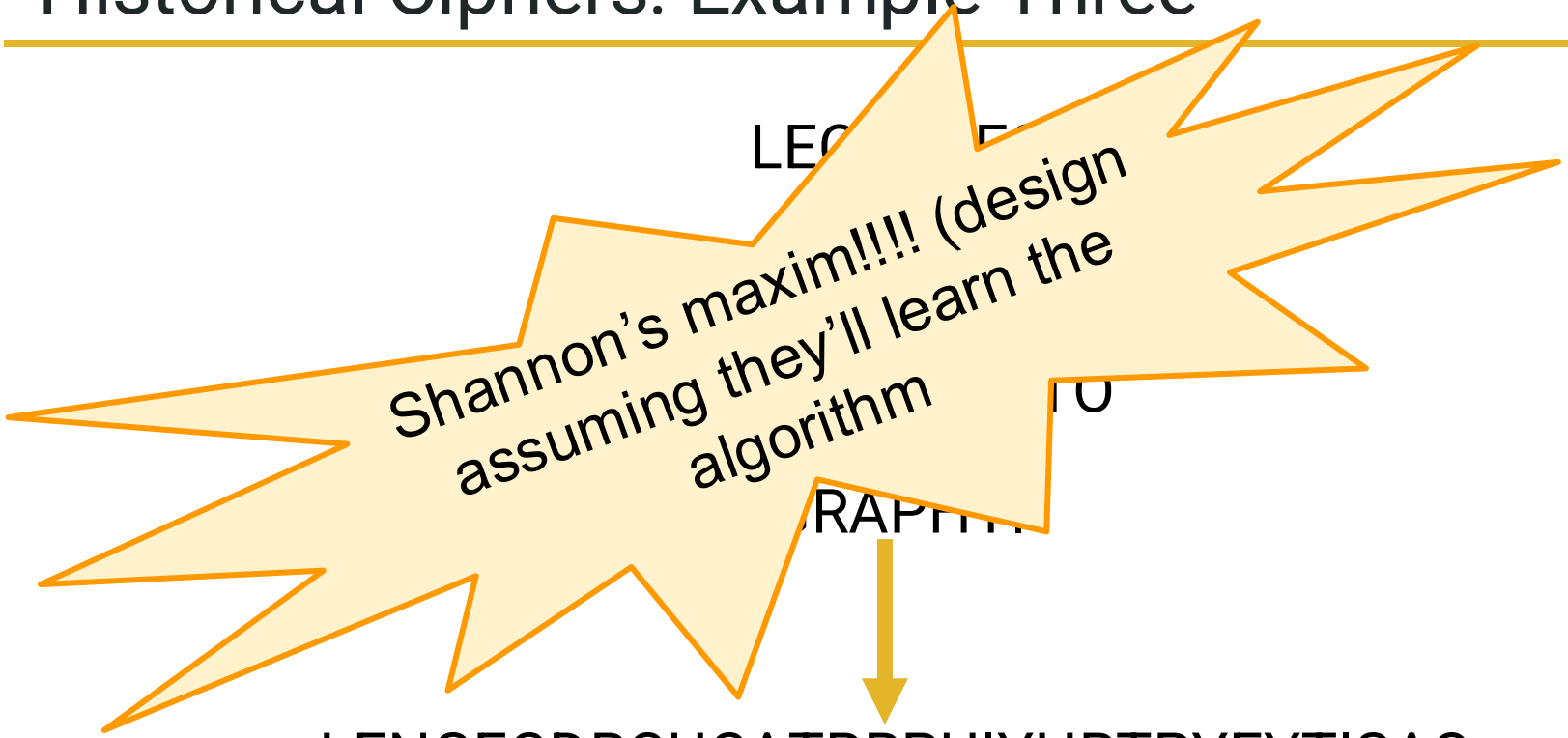
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Historical Ciphers: Example Three



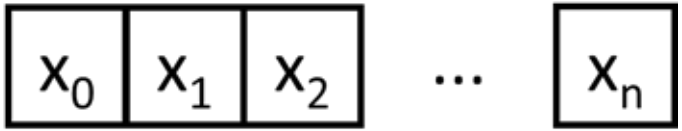
LENGECDRCUCATRRPUIYHRTPYEYTISAO

Shannon's Maxim and Kerkhoff's Principle Mean:

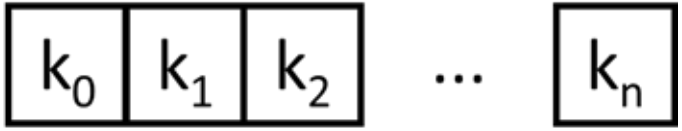
- Security shouldn't rely on the secrecy of the method
- Do use public algorithms with secret "keys"
- The adversaries target…is the key

Key: Easier to change a "short" key than your whole system. (e.g., Recovery)

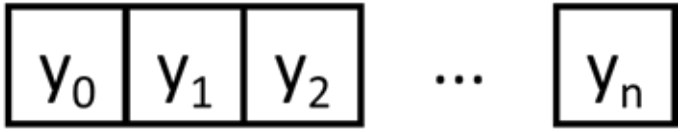
Unconditionally Secure: One-Time Pad

Message: 

$\oplus$

Key: 

=

Ciphertext: 

Rule: $y_i = x_i + k_i \pmod 2$

Provably Security for One-Time Pad

<Ciphertext is uniformly distributed independent of the plaintext distribution>

$x_i = 0$ with probability p ($x_i = 1$: 1-p), $k_i = 0$ with probability 0.5 ($k_i = 1$: 0.5), $y_i = 0$ with probability:

$$p(y_i = 0) = p(x_i = 0)\, p(k_i = 0) + p(x_i = 1)\, p(k_i = 1)$$

$$= 0.5p + 0.5(1-p)$$

$$= 0.5$$

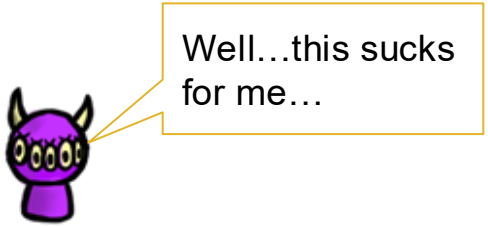
Provably Secure Con’t

Every ciphertext y can be decrypted **into every arbitrary plaintext** x using the key

k = yx

Consequently the ciphertext cannot contain any information about the plaintext

Encryption is “deniable”



What if it is a many-time pad?

Key: K

$Ciphertext_1 = message_1$ xor K = 2c1549100043130b1000290a1b

$Ciphertext_2 = message_2$ xor K = 3f16421617175203114c020b1c



Hmmm…what do I know these are made of…and definitely contain?

Many-time pad? Messages Lack True Randomness



C_1



C_2



$C_1 \oplus C_2$



M_2

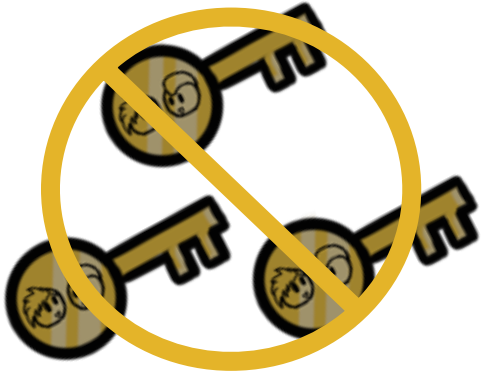


M_1

One-Time Pad - Conditions…

- Key as long as the message
- Key uniformly random
- Only used once





So…Cryptography?

- Simple substitution/transposition is computationally insecure
- One-Time Pad is inefficient over the secure channel

Goal: Securely communicate “a lot” of information on an insecure channel while requiring “limited” communication over a secure channel

Recap: A, B, C versus A and B and C

Substitution is insecure…

Transposition is insecure…

Key reuse using XOR (one-time pad) is insecure…

BUT

Repeat it often enough and it can be widely regarded as secure

Recap: A, B, C versus A and B and C

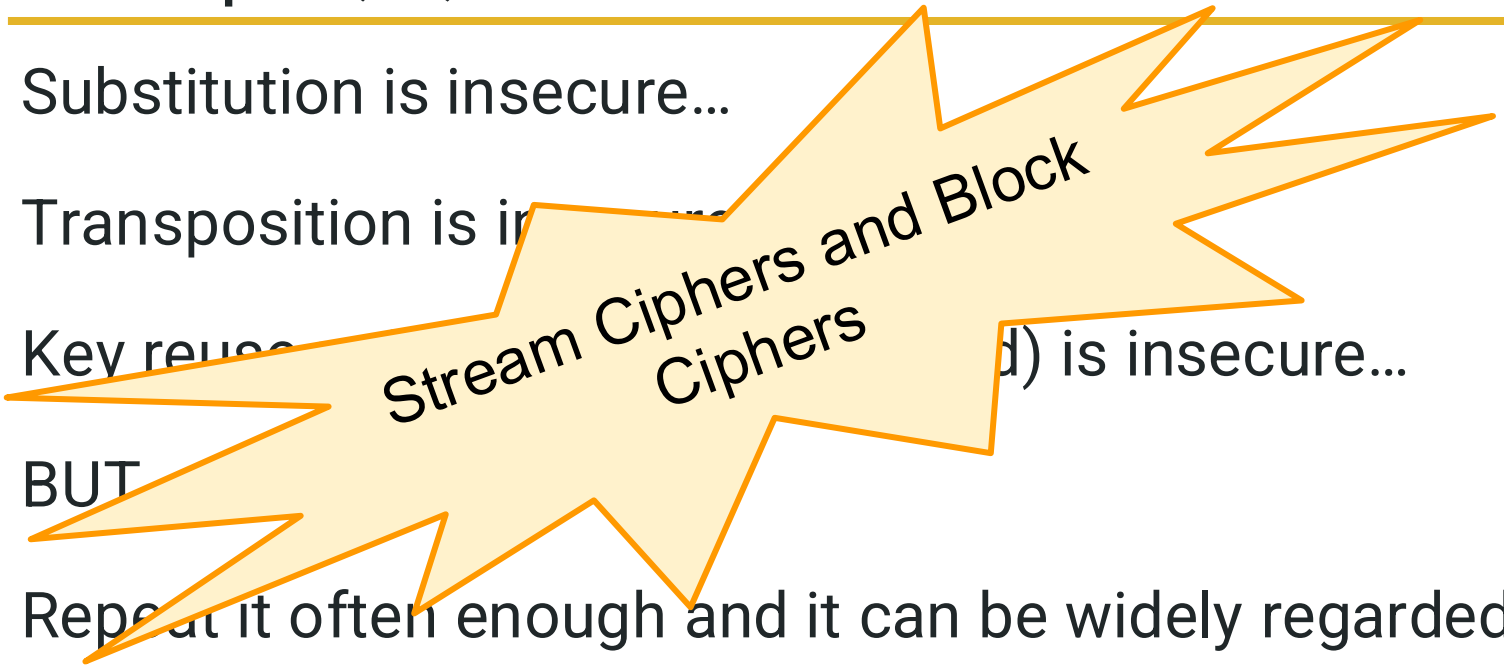
Substitution is insecure…

Transposition is in…

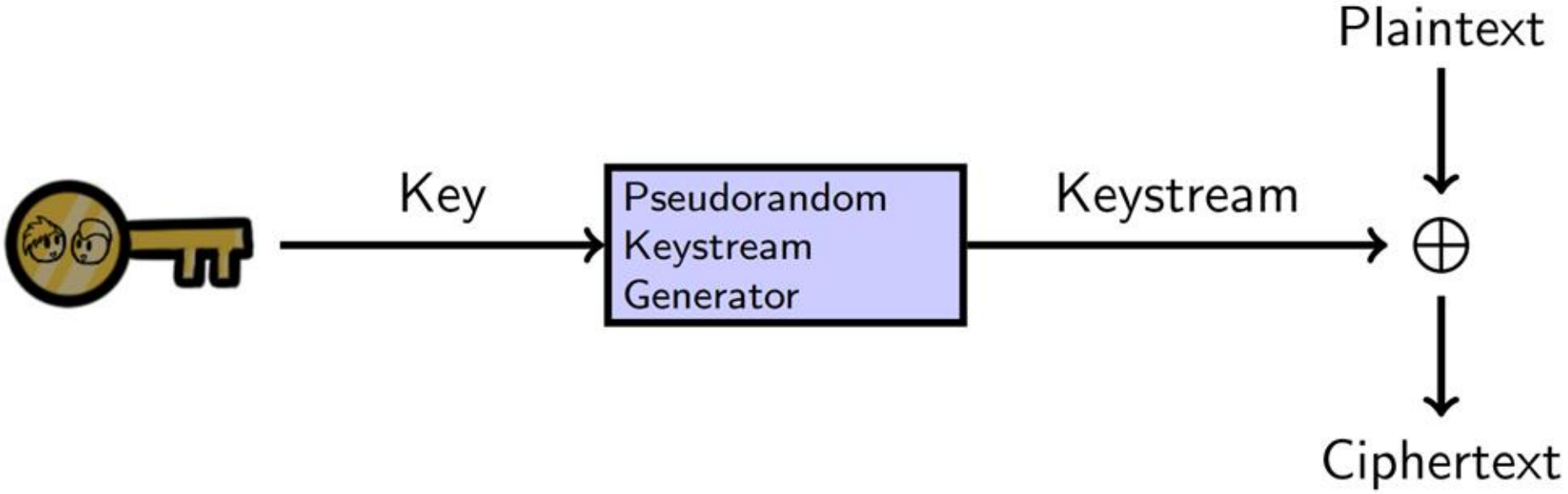
Key reuse…d) is insecure…

BUT

Repeat it often enough and it can be widely regarded as secure



Stream Cipher?

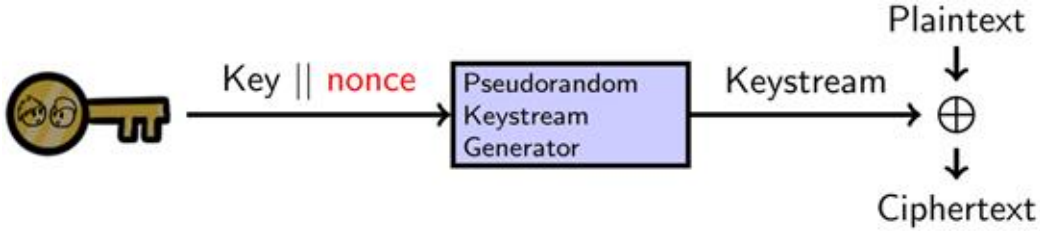


Fun(?) Facts:

- RC4 was the most common stream cipher on the Internet but deprecated.
- ChaCha increasingly popular (Chrome and Android), and SNOW3G in mobile phone networks.

Stream Ciphers Share Conditions with OTP

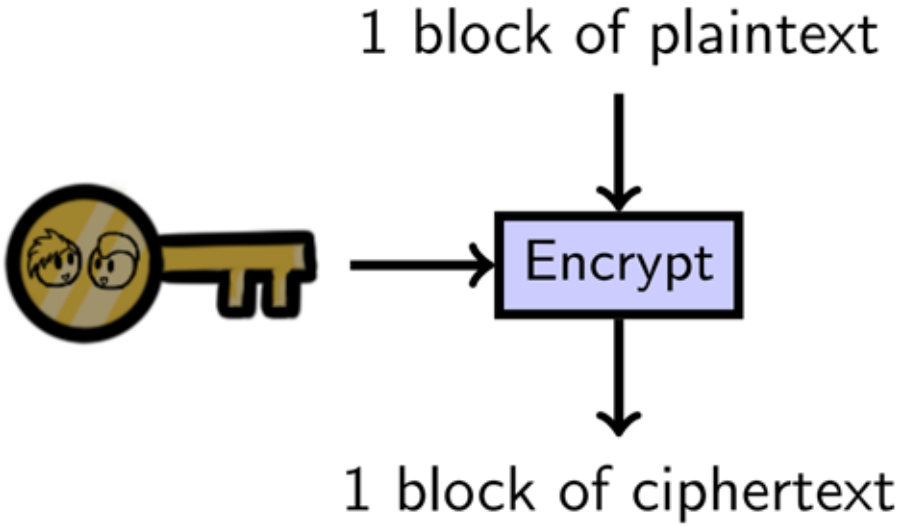
- Stream ciphers can be very fast
  - This is useful if you need to send a lot of data securely
- But they can be tricky to use correctly!
  - We saw the issues of re-using a key! (two-time pad)
  - Solution: concatenate key with nonce (we'll see more about nonces later)



Fun(?) Facts:

- WEP, PPTP are great examples of how not to use stream ciphers

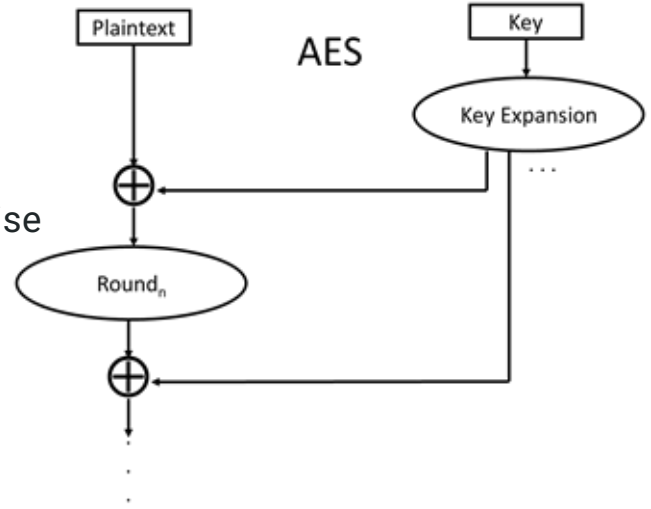
Bit by bit….do you have to?



Block ciphers!!!

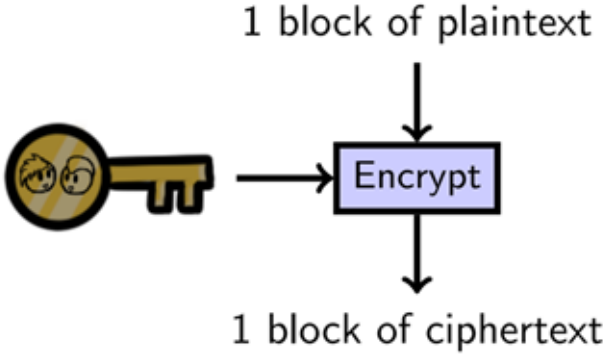
Block Ciphers

- Weakness of streams…one bit at a time?
  - What happens in a stream cipher if you change just one bit of the plaintext?
- Welcome, use of block ciphers
  - Block ciphers operate on the message one block at a time
  - Blocks are usually 64 or 128 bits long
- **AES,** the current standard
  - You better have a very…very good reason to choose otherwise

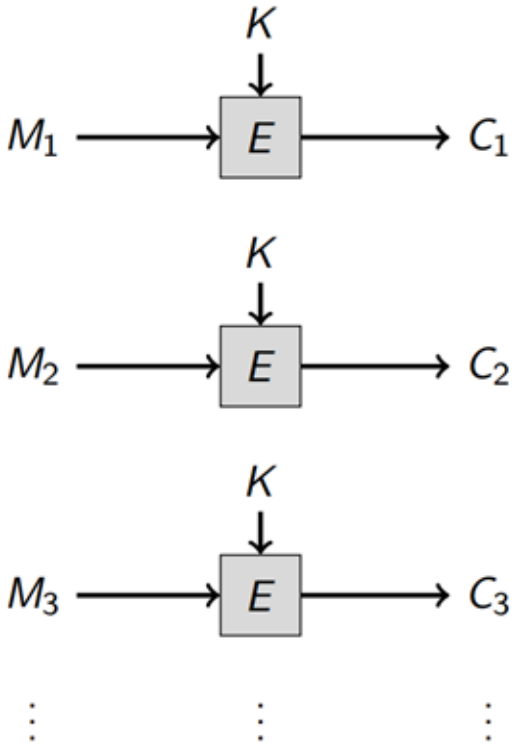


Two Catches with Block Ciphers

- Message is shorter than one block
  - padding
- Message is longer than a block
  - Modes of operation <new concept>

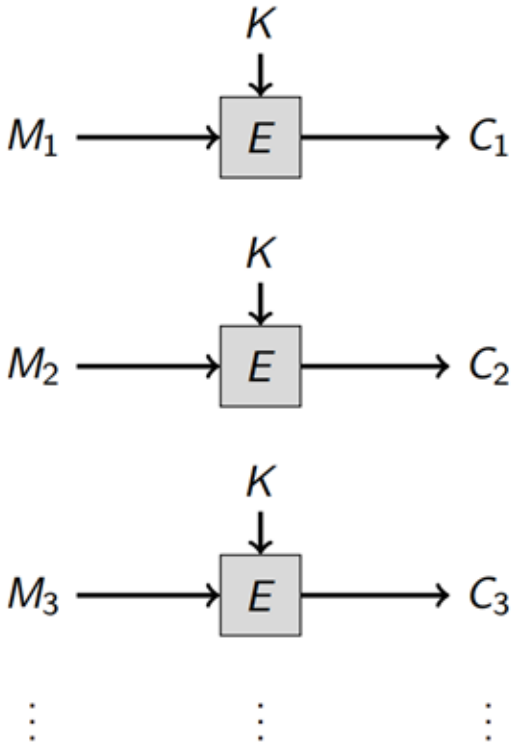


Block Ciphers and Modes of Operation: ECB Mode



- ECB: Electronic Code Book
- Encrypts each successive block separately

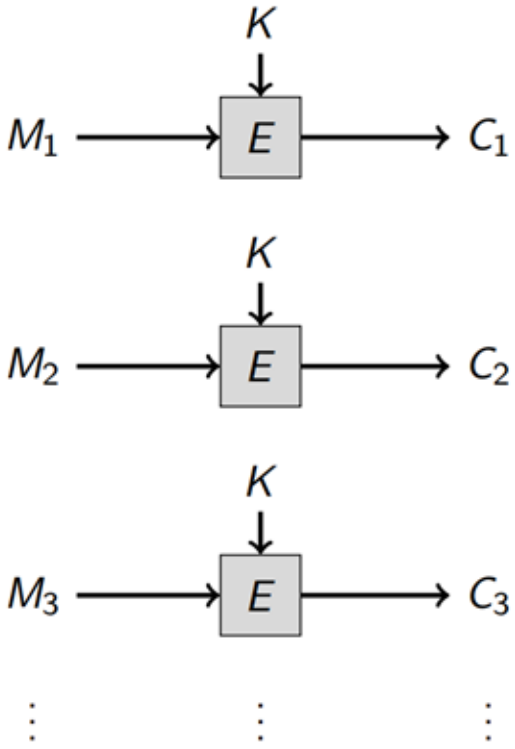
Block Ciphers and Modes of Operation: ECB Mode



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Q: What happens if the plaintext M has some blocks that are identical, $M_i = M_j$?

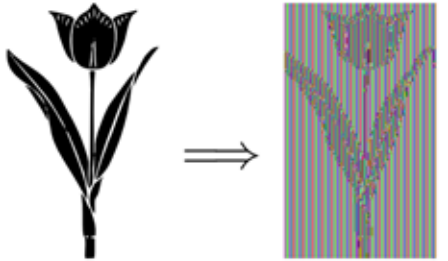
Block Ciphers and Modes of Operation: ECB Mode



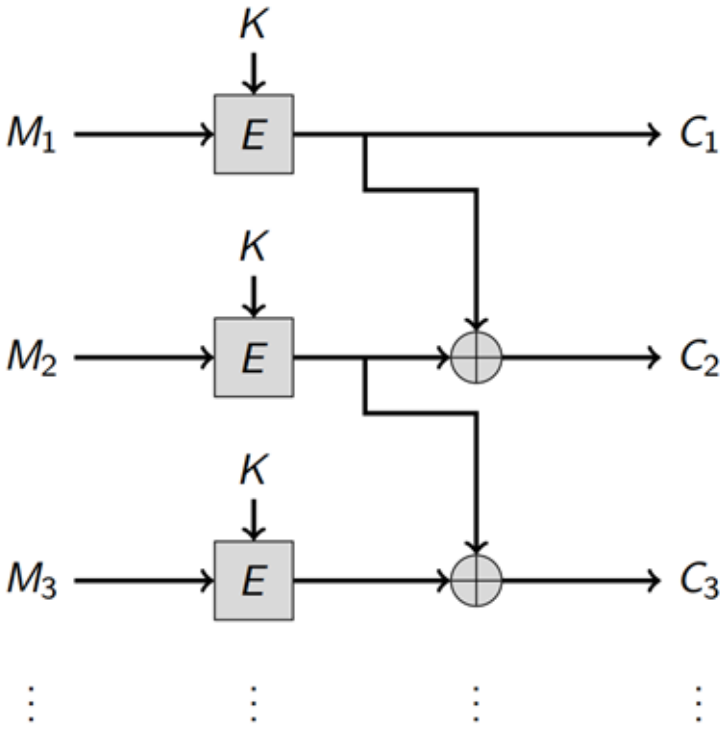
- ECB: Electronic Code Book
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Q: What happens if the plaintext M has some blocks that are identical, $M_i = M_j$?

A: $C_i = E_K(M_i)$, $C_j = E_K(M_j) \Rightarrow C_i = C_j$



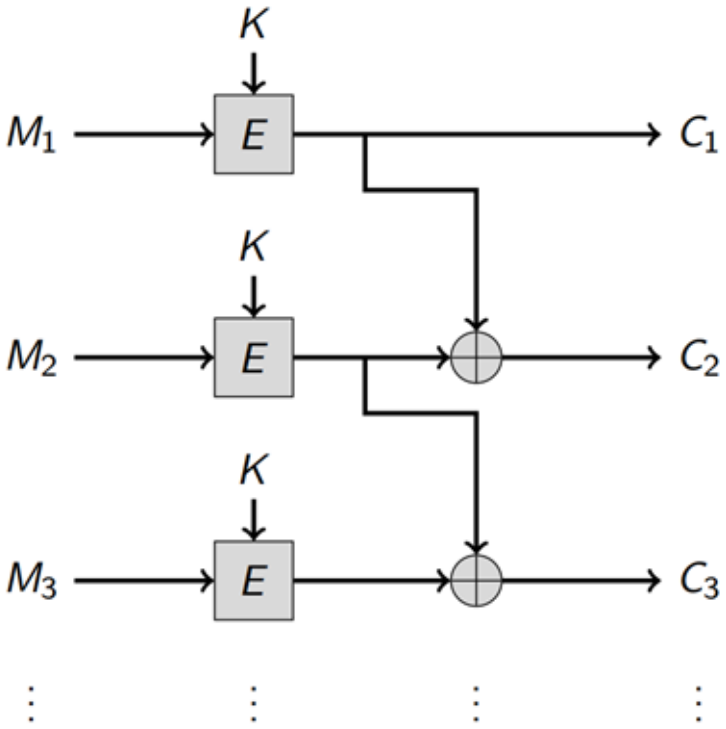
Attempt 1: Fixing ECB_1



- Provide “feedback” among different blocks, to avoid repeating patterns…

Q: Fix repeating patterns? Are there other issues?

Attempt 1: Fixing ECB_1

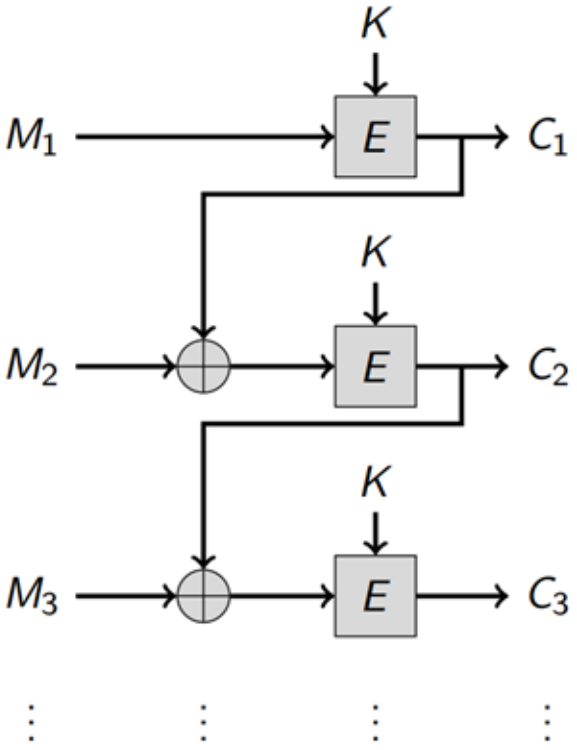


- Provide “feedback” among different blocks, to avoid repeating patterns…

Q: Fix repeating patterns? Are there other issues?

A: We can un-do the XOR if we get all the ciphertexts. This basically does not improve compared to ECB.

Attempt 2: ECB_2!!!

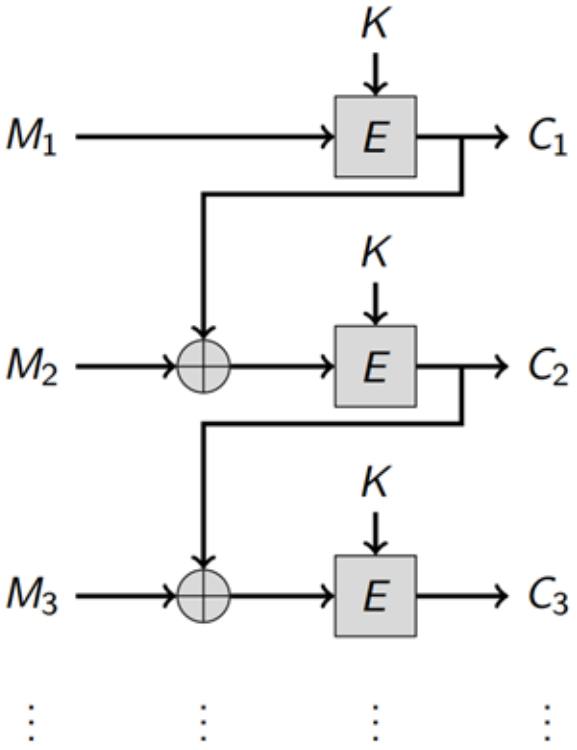


Q: Spot the difference?

Q: Is it fixed this time?

Q: Does this avoid repeating patterns among blocks?

Attempt 2: ECB_2!!!



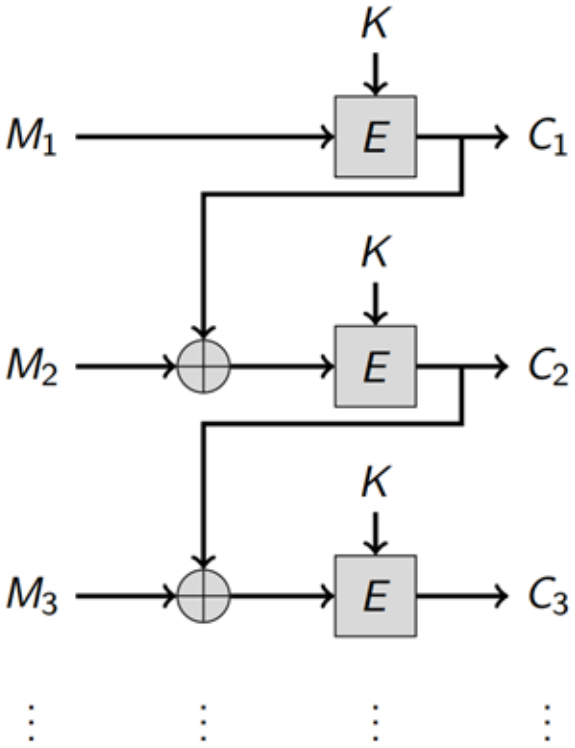
Q: Spot the difference?

Q: Is it fixed this time?

Q: Does this avoid repeating patterns among blocks?

Q: What would happen if we encrypt the message twice with the same key?

Attempt 2: ECB_2!!!



Q: Spot the difference?

Q: Is it fixed this time?

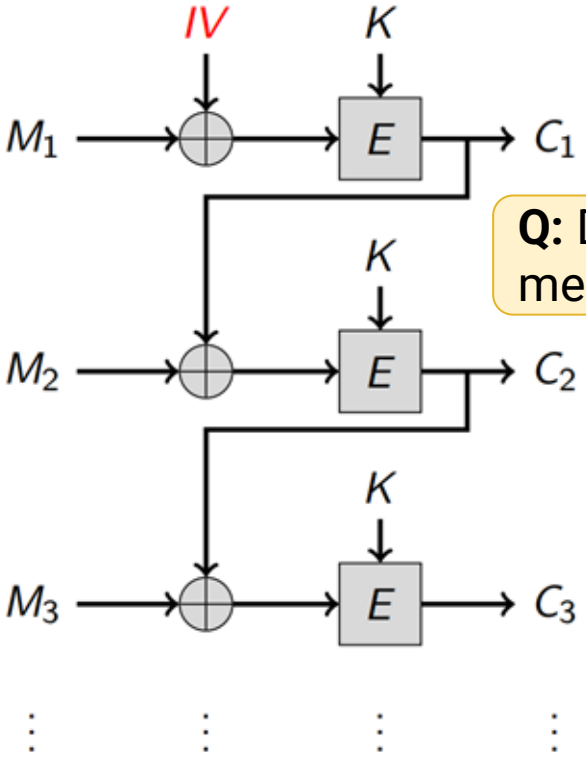
Q: Does this avoid repeating patterns among blocks?

Q: What would happen if we encrypt the message twice with the same key?

A: $C_1 = E_K(M), C_1' = E_K(M) \Rightarrow C_1 = C_1'$



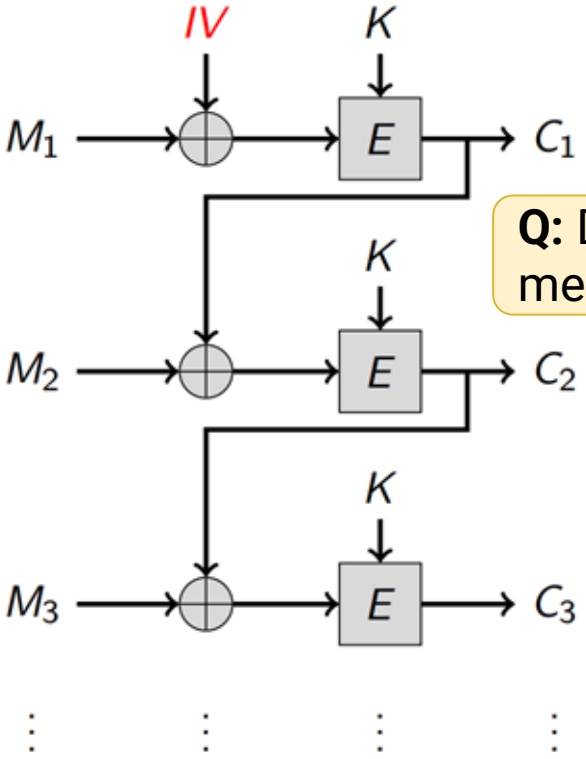
New Plan: CBC Mode



Q: Does this solve the issue of encrypting equal blocks?

Q: Does this solve the issue of encrypting equal messages/plaintexts?

New Plan: CBC Mode



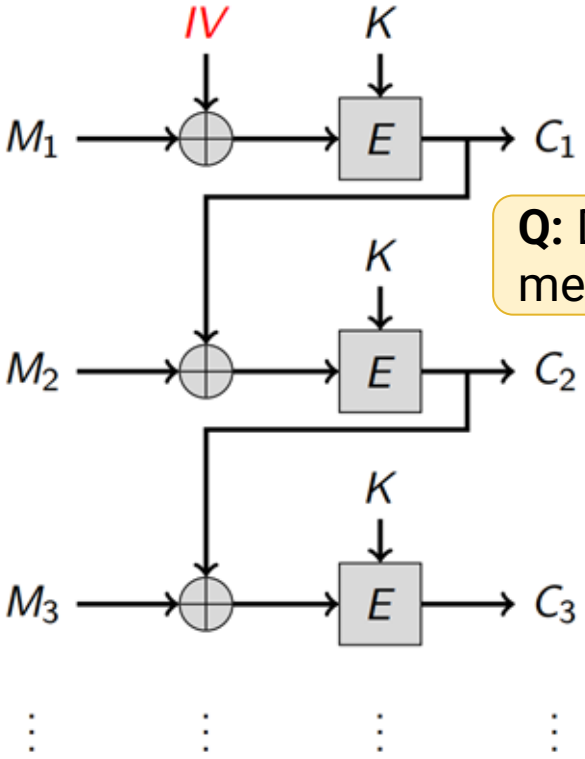
Q: Does this solve the issue of encrypting equal blocks?

Q: Does this solve the issue of encrypting equal messages/plaintexts?

A: Yes!!!



New Plan: CBC Mode



Q: Does this solve the issue of encrypting equal blocks?

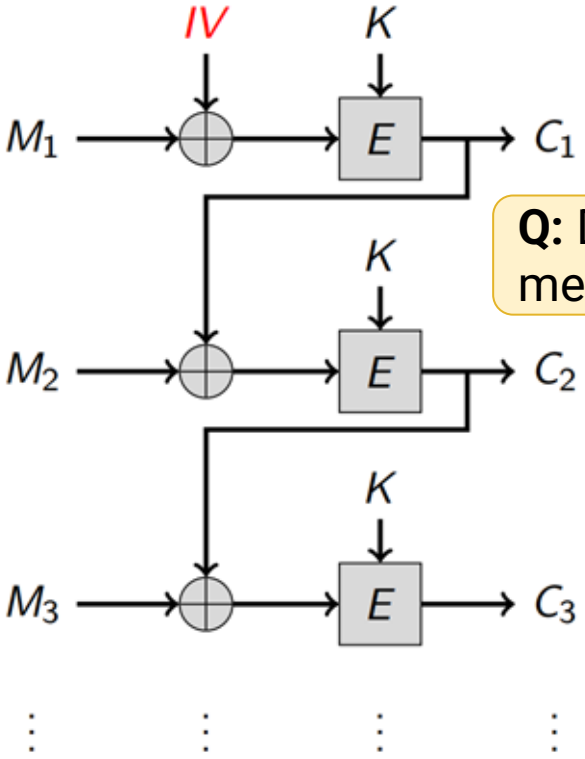
Q: Does this solve the issue of encrypting equal messages/plaintexts?

A: Yes!!!



Q: Can we share IV in the clear?

New Plan: CBC Mode



Q: Does this solve the issue of encrypting equal blocks?

Q: Does this solve the issue of encrypting equal messages/plaintexts?

A: Yes!!!



Q: Can we share IV in the clear?

A: Yes!!!

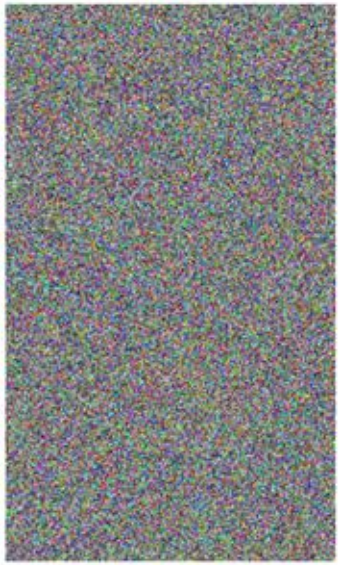


IV, an initialization vector, nonce, salt.

Modes of Operation Collection

- Cipher Block Chaining **(CBC)**, Counter **(CTR)**, and Galois Counter **(GCM)** modes
- Patterns in the plaintext are no longer exposed because these modes involve some kind of “feedback” among different blocks.
- But you need an **IV**





So…now what?

- How do Alice and Bob share the secret key?
  - Meet in person; diplomatic courier…
- In general this is very hard

Or, we invent new technology!!

Spoiler Alert: it's already been invented…

Tuesdayyyyyyyyyy

Until next time...